

# Chun-Yu (Echo) Shih

**Full-Stack Engineer (Frontend-focused) · React · TypeScript · End-to-End Product Development**

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## Summary

Full-stack engineer with a strong focus on frontend systems, specialising in **React, Next.js, and TypeScript**, and comfortable owning the backend — **FastAPI, event-driven services, and CI/CD** — end to end.

Experienced building production applications from the UI and complex state down to backend APIs, data modeling, testing across the stack, and the deployment pipeline. Recent work spans an **event-driven modular-monolith platform**, a **Supabase-backed multi-tenant web app**, and a **Flutter mobile app shipped to both app stores**.

Particularly interested in **local-first software and data-intensive interfaces**, with an emphasis on building reliable, secure, and user-focused products.

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## Engineering Impact

- **Built and operate production systems end-to-end** — frontend (React / Next.js), backend APIs (FastAPI), data layer, and deployment.
- **Architected an event-driven modular monolith** with bounded modules communicating over a **RabbitMQ message bus**, backed by **ArangoDB** (multi-model).
- **Designed role- and policy-based access control** with secure authentication (JWT, Argon2), and hardened data access via **Postgres Row-Level Security**.
- **Engineered a build-once, promote-by-digest CI/CD pipeline** (GitHub Actions → GHCR) that eliminated redundant rebuilds and guarantees production ships the exact image validated in staging.
- **Shipped local-first applications** with encrypted on-device persistence (**SQLCIPHER / SQLite**) and offline-capable data flows.

- **Delivered AI-powered production features** integrating OpenAI / ChatGPT APIs within live products.
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## Experience

### Lockerbie Technology

#### Full-Stack Engineer — Internal Platform Development

Taipei, Taiwan · Dec 2024 – Present

- Building an internal team project-management platform end-to-end as one of two engineers, owning both the **Next.js 15 / React 19 (TypeScript)** frontend and the **FastAPI / Python 3.12** backend.
- Architected an **event-driven modular monolith** — bounded modules for auth, organisation, user, project, and project management communicating asynchronously over a **RabbitMQ message bus (aio-pika)** with a dedicated consumer service for background processing.
- Modeled the data layer on **ArangoDB (multi-model)**, representing organisational and project relationships as connected graph/document data.
- Designed **role- and policy-based access control** — a role→permission/capability model enforced across backend modules and mirrored in the frontend — with secure auth via **JWT (python-jose)** and **Argon2 password hashing**.
- Built the **Next.js frontend** (MUI, TanStack Query, Zustand) delivering **dual client/vendor viewpoints** where a single user can act as either, plus a **superadmin dashboard** that visualises project node/edge relationships with **force-graph / Three.js** for at-a-glance oversight and fast cleanup of test data.
- Established **full-stack quality gates** — Vitest + React Testing Library + **Playwright E2E** + MSW (frontend), **pytest / pytest-asyncio** (backend), with **mypy (strict)** and **Ruff**.
- Re-architected the deploy pipeline to **build-once, promote-by-digest** (GitHub Actions → **GHCR**): images build a single time on staging (multi-arch amd64/arm64), then production and tagged (v\*) releases **re-tag the exact digest** instead of rebuilding — cutting a release path that previously rebuilt **up to 3×** down to one build, reducing CI time and GitHub storage while guaranteeing prod ships the byte-identical image validated in staging.
- Hardened the release flow with **least-privilege job permissions, buildx layer caching, automated cache cleanup, and**

**conventional-commit / tag-driven versioning**, plus auto-generated staging→production promotion PRs.

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## SPIN.FASHION by Lablaco

**Software Engineer · Founding Partner**

Singapore · Remote · Aug 2024 – Oct 2024

- Resolved a **critical macOS Electron release blocker** — fixing code-signing, **Apple notarization**, and **S3-based auto-update** — to unblock desktop distribution of the in-store app (React / Redux-Saga).
  - Authored a **release runbook** for the signed and notarized macOS + Windows builds, adopted as the team's deployment standard.
  - Improved **React frontend performance and stability** across the Electron renderer.
  - Delivered remotely across time zones as a founding partner on a small distributed team.
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## Find Recruiter (HR Tech Startup)

**Software Engineer**

Taipei, Taiwan · Aug 2022 – May 2024

- One of two engineers on a **B2B recruitment / applicant-tracking platform** (React 18, ~150k LOC), leading frontend development across employer and recruiter workflows.
  - Built and shipped **multiple OpenAI / ChatGPT-powered features** — a **JD Generator**, AI interview-plan generation, and a CV standardiser — orchestrating server-side LLM calls from the frontend.
  - Drove an **incremental JavaScript → TypeScript migration** of a large production codebase, reducing runtime bugs by ~30%.
  - Developed **100+ reusable UI components** with **Redux Toolkit + redux-saga** state management and **Firebase-backed realtime chat and notifications**.
  - Shipped a **5-language internationalised** product deployed via GitHub Actions CI/CD to **GCP Cloud Run and AWS**.
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## Independent Engineering Work

### Software Engineer

Remote · 2020 – Present

- Built full-stack applications across fintech, e-commerce, and research domains, adapting system design to varied product requirements.
  - Built the **Next.js / React frontend for a Bitcoin coin-selection demo wallet** (*echology*), in collaboration with a **BDK core maintainer** — implementing UTXO coin-control selection, spend-scenario and fee-rate configuration, and surfacing coin-selection metrics (waste, fee, feerate deviation).
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## Founder & Product Engineering

### Cosmora — Salon Management Platform

#### Founder · Full-Stack Product Engineer

2024 – Present

*React · TypeScript · Supabase (Postgres) · Tauri (desktop packaging)*

- Built a **bilingual, multi-tenant salon-management application** solo (React 19 + TypeScript), packaged for desktop with Tauri and shipped as a macOS build.
- Designed a **domain-driven frontend architecture** (7 domains, strict `api` → `provider` import boundaries) with the boundaries **enforced at test time** by a custom static-analysis contract.
- Hardened data security with **Postgres Row-Level Security** scoped to org membership, a **per-member JSONB permission model**, and **SECURITY DEFINER** RPCs on Supabase.
- Implemented an **offline-capable read layer** (React Query cache hydrated from local storage) and secure staff provisioning via a **Supabase Edge Function**.
- Set up **Jest + React Testing Library with coverage gates** and **semantic-release CI** (automated versioning from conventional commits).
- Now designing the **next-generation version (SalonOS)** — product spec, financial engine, and visit-logging flows in active design.

## Runway — Personal Financial Runway App

Founder · Mobile Product Engineer

2026 – Present

*Flutter · Dart · Riverpod · Drift (SQLite) · SQLCipher*

- Built a **near-production cross-platform mobile app** (iOS + Android) that computes how long a person's savings will last — solo, in **Flutter / Dart**.
  - Structured a **Melos monorepo with Clean Architecture / DDD** — a pure-Dart domain layer with compile-time-enforced package boundaries and dedicated logic engines for burn rate, loan amortization, and subscription normalization.
  - Implemented **AES-256 encrypted local-first storage** (Drift + SQLCipher, key held in device secure storage) with versioned schema migrations and screenshot/recording blocking.
  - Shipped **in-app purchases (RevenueCat)**, **7-language localization**, ~100 unit/integration tests, and **CI/CD releasing signed builds to TestFlight and Google Play** on version tags.
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## Skills

### Frontend

React · Next.js · TypeScript · MUI · TailwindCSS

### Mobile

Flutter · Dart · Riverpod · SQLCipher

### Backend

FastAPI · Python · REST APIs · Pydantic · Node.js · Supabase

### State & Data

TanStack Query · Zustand · Redux · Redux-Saga · ArangoDB · SQLite · Drift · Firebase · RabbitMQ

### Testing

Vitest · Playwright · React Testing Library · MSW · pytest · Jest

## Infrastructure & DevOps

Docker · Docker Compose · GHCR · GitHub Actions · buildx · Linux (Ubuntu) · AWS · Vercel

## Build & Automation

Makefile · Poetry · semantic versioning (conventional commits)

## Languages & Runtimes

Python · Electron · Tauri (desktop packaging)

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## Education

### B.A. Italian Language and Culture

Fu Jen Catholic University · Taipei · 2019

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## Certifications

CS50 Cybersecurity — Harvard / edX (*In Progress*)

Data Science & AI with Python — Preface

LangChain for JavaScript Developers

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## Engineering Interests

Local-First Software · Data-Intensive Interfaces · Developer Tooling · Product-Focused Engineering